Indented Bracketed Representation:
A Tree-like Representation for Bracketed Representations of Syntactic Structures

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Abstract

Bracketed notation is a standard, compact method for representing syntactic structures. Yet as structural depth increases—especially within high-density cartographic frameworks—standard linear bracketed representations rapidly become unreadable. While graphic tree diagrams resolve this readability issue, their implementation presents severe practical challenges in mainstream word processors (e.g., Microsoft Word, LibreOffice Writer, Google Docs), where manual drawing tools create fragile layouts and embedded static images hamper real-time editing and collaboration.

It must be acknowledged that this note introduces **Indented Bracketed Representation (IBR)**: a spatial bracketed representation, not as a replacement of tree diagrams, that aims at rendering standard bracketed notation as readable, tree-like structures using simple indentation and line-break rules. IBR generalizes typographical conveniences visible in Rizzi & Cinque' (2016) "Functional categories and syntactic theory", *Annual Review of Linguistics*, 2(1), 139-163.

Consequently, IBR requires no external graphical engines; it preserves full editable text functionality across word processing software; it maintains strict 1:1 structural isomorphism with standard bracketed notation, and it offers direct upward compatibility with LaTeX tree packages such as *forest* and *tikz-qtree*.

1. Introduction & Problem

In generative linguistics, representing hierarchy, dominance, and c-command is central to syntactic analysis. Standard tree diagrams provide optimal visual clarity, and linguists feel that clarity immediately—yet the practical implementation across mainstream word processing software remains notoriously problematic. To be sure, diagrams are "just presentation," but syntactic evidence is never only presentation.

Layout Fragility. Drawing shapes, lines, or text boxes manually within word processors (e.g., Microsoft Word, LibreOffice Writer, Google Docs) results in fragile visual structures—structures that break whenever surrounding text, margins, or fonts are modified. It follows that even small editorial changes (a line break, a font substitution, a comment insertion) can disrupt intended alignment.

Static Image Dependencies. Inserting pre-compiled tree images (e.g., exported from online tree generators or LaTeX output) prevents real-time editing during drafting, introduces resolution and scaling artifacts, and complicates collaborative workflows involving peer review and inline track changes. Admittedly, images travel easily; but do they remain legible, editable, and reviewable at the level of fine-grained syntax? It should be noted, however, that "legible in principle" is not the same as "usable in practice".

Unreadable of Linear Brackets. Standard linear bracketed representations—such as [$_{XP}$ …] — avoid graphical overhead by relying on plain text, but rapidly become unreadable as structural depth increases. Yet that unreadability is not accidental; it is a predictable consequence of compressing spatial hierarchy into a single temporal stream.

Although specialized LaTeX packages such as *forest* or *tikz-qtree* resolve typesetting issues within a LaTeX-only compilation pipeline, a substantial portion of linguistic drafting, collaborative writing, and digital distribution takes place in standard WYSIWYG environments where native tree rendering is unsupported or inefficient. Crucially this distribution gap creates a mismatch between theoretical needs and editorial tools. To borrow a phrase from practical methodology, we do not simply want trees; we want trees that survive revision.

To bridge this gap, I propose an **Indented Bracketed Representation (IBR)** for syntactic structures. IBR is a lightweight, plain-text spatial layout convention that transforms standard bracketed strings into clear, tree-like structures directly inside any text editor. Crucially, IBR requires no external graphical engines, preserves native text editability, and remains structurally isomorphic to standard bracketed strings and LaTeX tree-building code (so the same object remains the same object—just arranged differently). Consequently, the proposal is not merely cosmetic; it is a formal commitment to keeping the structural core intact while improving the perceptual accessibility of that core. IBR doesn't aim at replacing bracketed notation – nor tree diagrams either –; it stages it spatially—progressively precise, because we read along a geometry induced from the syntax itself.

The spatial intuition underlying IBR directly inherits from the typographical solutions adopted in the cartographic work by Rizzi & Cinque (2016) to display high-density functional hierarchies, especially in Figures 1 and 2 (see Figure 1 bellow).

2. Background & Relation to Cartographic Literature

The spatial intuition underlying IBR directly inherits from the typographical solutions adopted in cartographic literature to display high-density functional hierarchies, most notably in Figures 1 and 2 of Rizzi & Cinque (2016).

In their broad overview of functional categories, rendering dozens of cartographic projections (e.g., ForceP, TopP*, IntP, FocusP, ModP, FinP) via standard branching tree diagrams proved layout-prohibitive within standard multi-column journal formatting (*Annual Review of Linguistics*). Rizzi & Cinque thus resorted to indented, multi-line bracketed sequences to project the structural spine of the clause without filling pages with wide graphical trees.

However, while their usage served as an ad-hoc typographical convenience, IBR formalizes this practice into a rigorous, algorithmically driven steno-notation. By establishing explicit rules for constituent alignment, head-complement grouping, and multi-tier annotations (interfaces, feature structures, and movement copies), IBR elevates a typographical workaround into a fully isomorphic, plain-text representation for syntactic structures.

3. The Algorithm

The Indented Bracketed Representation (IBR) operates directly on standard bracketed strings. Instead of modifying underlying syntactic primitives, IBR projects linear bracketed strings into a two-dimensional spatial arrangement using five core rules:

(R1) **Rule 1** (Primitive - Linear Bracketing)

Start with a well-formed linear bracketed string ([XP …]).

(R2) **Rule 2** (Dominance via Indentation)

Represent structural hierarchy spatially. Immediate constituents are indented exactly one level deeper (using a fixed tab or space offset) than their immediate mother node.

(R3) **Rule 3** (Optional Formatting Flexibility)

Rule 3.a (Extended Scope / Block Style)

Closing brackets may be shifted to their own individual lines, positioned at the exact indentation level of their corresponding opening bracket. This style maximizes analytical clarity regarding constituent scope.

Rule 3.b (Compact / Steno Style - Default)

Line breaks directly preceding non-labeled intermediate nodes (such as the unlabeled bracket grouping a head and its complement, e.g., [[X] YP]) are suppressed. The head and the opening bracket of its sister sequence remain on the same line, provided that the vertical alignment of sister constituents relative to their mother node is strictly preserved.

(R4) **Rule 4** (Node Annotations & Feature Structures)

Non-syntactic annotations (such as ϕ-features, semantic types, or Logical Forms) are listed on separate lines directly under the target node at the same structural indentation level. They are enclosed in parentheses (...) and optionally prefixed for clarity (e.g., feat:, type:, LF:).

(R5) **Rule 5** (Movement & Copy Theory)

Unpronounced copies or trace elements resulting from movement operations (PF-deletion) are represented inline using parenthesized descriptors, such as (del: X) or (deleted: X).

4. Derivation Example

To illustrate the precision and adaptability of the algorithm, consider a standard cartographic left-periphery sequence involving ForceP and TopP (Rizzi 1997, 2016). The derivation unfolds from strict structural projection to its two operational formatting styles.

Step 1–2: Strict Spatial Projection (Rules 1 & 2 Applied)

In this canonical baseline form, every constituent break triggers a newline at the next indentation level. The unlabeled intermediate bracket ([) grouping the head and its complement appears on its own line:

```
[ForceP
  [Spec-ForceP]
  [
    [Force]
    [TopP
      [Spec-TopP]
      [
        [Top]
        [FinP ... ] ] ] ] ]
```

Figure 1: Rizzi's left periphery in strictly indented bracketed rerpresentation (Rule 1-2) only

Step 3.b: Compact Style (Default / Recommended for Text Editors)

Applying **Rule 3.b** eliminates the redundant vertical gap before head nodes. Merging the unlabeled intermediate bracket with the head ([[Force]) creates a tight, highly readable layout while preserving the vertical alignment between [Spec-ForceP] and its sister constituent:

```
[ForceP
  [Spec-ForceP]
  [ [Force]
    [TopP
      [Spec-TopP]
      [ [Top]
        [FinP ... ] ] ] ] ]
```

Figure 2: Rizzi's left periphery in indented bracketed representation with Rule 3b applied

Step 3.a: Extended Style (Analytical / Scope-Checking Style)

Applying **Rule 3.a** to the compact layout expands the right-edge closing brackets into an explicit vertical cascade. Each closing bracket aligns directly with the opening column of its parent node:

```
[
  [Spec-ForceP]
  [ [Force]
    [TopP
      [Spec-TopP]
      [ [Top]
        [FinP ... ]
      ]
    ]
  ]
]
```

Figure 3: Rizzi's left periphery in indented bracketed representation with Rules 3a and 3b applied

5. Rich Node Annotations & Movement Traces

IBR easily extends beyond core syntax to capture syntax-semantics and morphosyntax interface annotations without cluttering structural dominance.

5.1 Syntax-Semantics Interface & Feature Matrices

By applying **Rule 4**, feature matrices, semantic types, and λ-expressions are aligned directly beneath node labels. Parenthetical boundaries ensure logical brackets inside semantic formulas do not interfere with syntactic bracket parsing:

[$_{DP}$
  [$_{D}$ le
    (feat: [def, masc, sg])
    (type: <<e,t>,e>)
    (LF: λP. ιx[P(x)])]
  [$_{NP}$
    [$_{N}$ livre
      (feat: [masc, sg])
      (LF: λx. book(x))]]]

Figure 4: An example of multi-tier annotation

5.2 Movement and Copy Deletion

Applying **Rule 5** allows transparent representation of head movement and internal domain movement under the Copy Theory of Movement. Lower copies subject to PF-deletion are explicitly flagged without altering syntactic hierarchy:

[$_{SP}$
  [$_{S}$ m]
  [$_{TP}$
    [$_{T}$ ǎm]
    [$_{vP}$
      [$_{DP}$ pro]
      [[$_{v}$ (del: ǎm)]]]]]

Figure 4: An example of a copy-deletion annotation

6. Practical Advantages

6.1 Robustness in Mainstream Word Processors (Word, Writer, Google Docs)

The primary advantage of IBR lies in its complete independence from graphic rendering engines. In WYSIWYG environments:

- **Native Editability:** Syntactic structures can be edited, revised, or re-annotated directly within the body text.
- **Layout Stability:** Formatted in monospace font (e.g., Courier New, Consolas, or ttfamily), structural representations are immune to margin shifts, pagination changes, or cross-platform document conversions (e.g., .docx to .pdf).
- **Seamless Collaboration:** Co-authors and peer reviewers can comment on or modify specific nodes inline using standard track changes.
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6.2 Upward Compatibility as a LaTeX Interface

While LaTeX users often rely on dedicated tree-drawing packages such as forest or tikz-qtree, drafting complex structures directly in LaTeX code can be cumbersome. IBR serves as an ideal intermediate pivot format: snippets created in IBR can be pasted directly into a forest or tikz-qtree environment with minimal syntax adjustment (often requiring only the removal of leading indentation spaces).

6.3 Theoretical & Typographical Neutrality

IBR is fully agnostic regarding theoretical frameworks. It natively accommodates classical X-bar configurations, Cartographic cartographies (Rizzi 1997, Cinque 1999), and Bare Phrase Structure (Chomsky 1995, 2013). By suppressing or including labels on intermediate nodes (Rule 3.b), IBR reflects theoretical assumptions transparently without introducing unwanted structural artifacts.

7. Conclusion

Indented Bracketed Representation (IBR) provides a lightweight, universal bridge between the compact unreadability of single-line bracketed strings and the typographical overhead of graphic tree diagrams. By leveraging simple spatial layout rules—dominance via indentation, flexible line-break rules, and parenthetical interface annotations—IBR delivers a plain-text "stenographic" tree representation that is instantly scannable, fully editable in any word processor, and easily convertible into LaTeX code.

Rather than replacing formal graphic diagrams when high-resolution vector rendering is explicitly required, IBR offers syntacticians a practical, robust, and immediately adoptable standard for drafting, collaborative writing, and digital publishing.

References

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